Causation and Research Design

August 29, 2022

Theory Activity

In Groups of 4...

- Read the following statement: "It has frequently been observed that students coming into a large lecture hall will tend to fill up the rear of the hall first."

- Then:
  1. Come up with an explanation for this observation
  2. *Generalize* the theory- that is, come up with the most general abstract theory you can produce that still has the original observation as a consequence
  3. Come up with an *alternative* explanation for the observation
  4. Generalize that alternative explanation
  5. Derive a new, testable hypothesis from each theory. Or, derive one *critical test* to rule out one of the theories

Questions

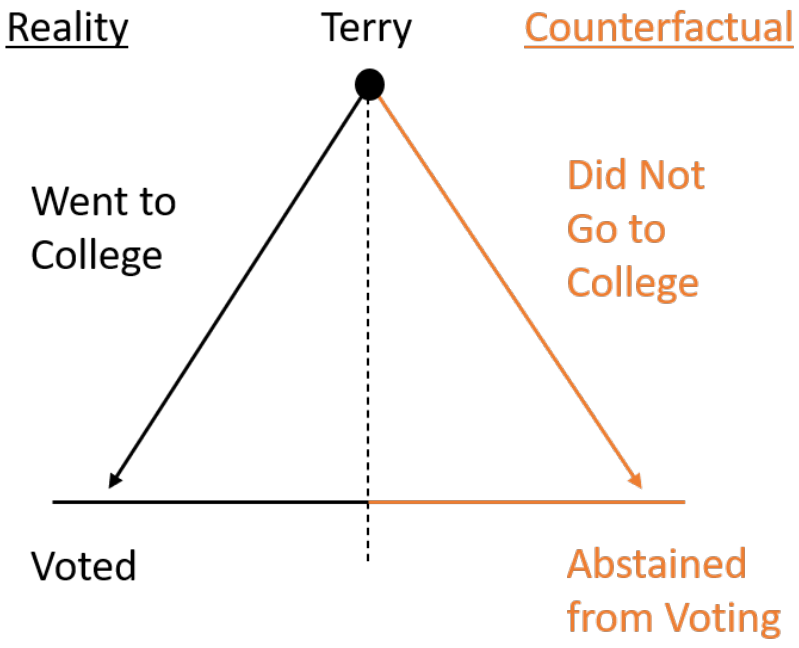
- Normative- What should the world look like?
- Empirical- Questions about how the world is.
  1. Descriptive- What does the world look like?
  2. Relational- Are a set of phenomena related to each other?
  3. **Causal- Why does the world look the way it does?**

Causality is difficult to establish in social science. Why?

Causation

Counterfactual- how the world would look if something did NOT happen

Observe: Terry is college educated and voted in the last election. Did college education *cause* Terry to vote?



Fundamental Problem of Causal Inference

To know if **X** <u>causes</u> **Y**, we must observe what happened in the presence of **X** AND what would have happened in the absence of **X**, which is impossible.

Fundamental Problem of Causal Inference

To know if **X** causes **Y**, we must observe what happened in the presence of **X** AND what would have happened in the absence of **X**, which is impossible.

This is especially difficult in social science because we live in a probabilistic world

- Deterministic relationship- if some cause occurs, then the effect will occur with certainty
- Probabilistic relationship- if some cause occurs, then there is a change in the probability that an effect occurs

Necessary Elements of Causal Relationships

1. There is <u>covariation</u> between **X** and **Y**
   - **X** and **Y** correlate
2. There is a plausible <u>causal mechanism</u>
   - Reasonable theory
3. **X** causes **Y**, <u>not the other way around</u> (temporal order)
   - Relationship is not "endogenous" (a.k.a. reverse causation)
4. <u>Confounders</u> are ruled out or controlled for
   - Relationship is not "spurious"

1. Covariation

Changes in one variable correspond to changes in another variable

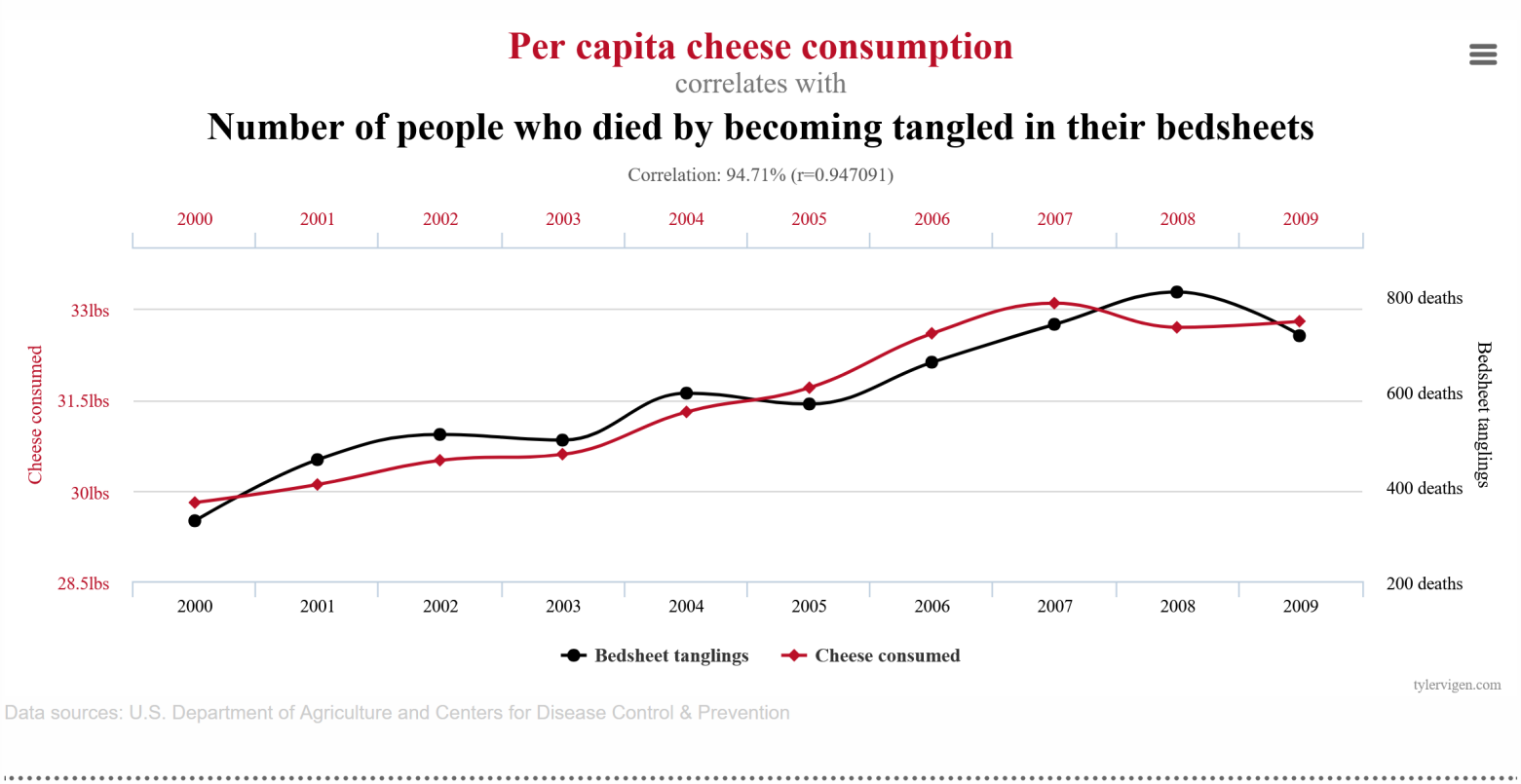
Common measures of association:

- Chi-square
- Gamma
- Pearson's R (Correlation)

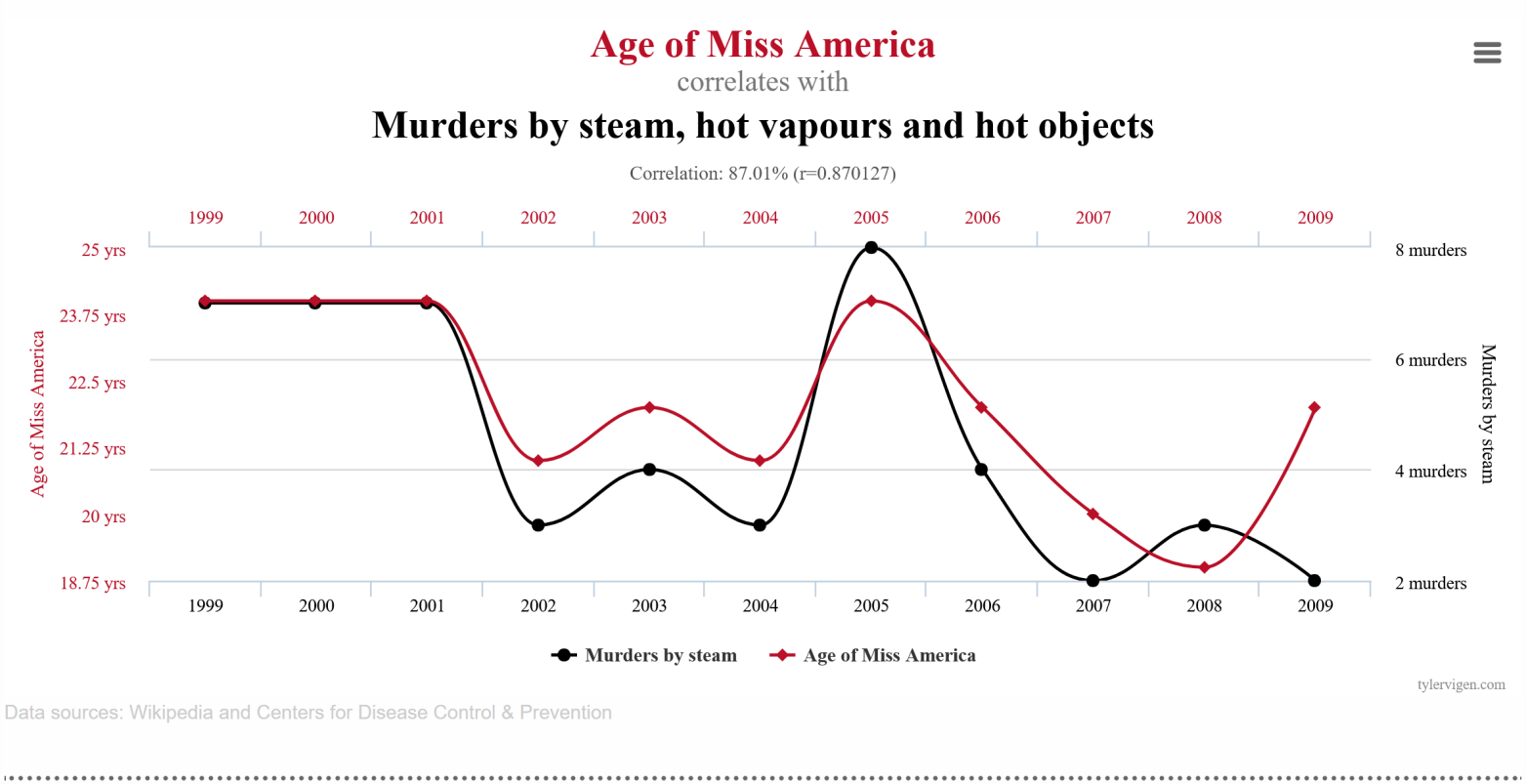
Note, however, that the (social) world is often multicausal

- Bivariate- two variables (e.g., one cause of one effect)
- Multivariate- multiple variables (e.g., multiple causes of one effect)

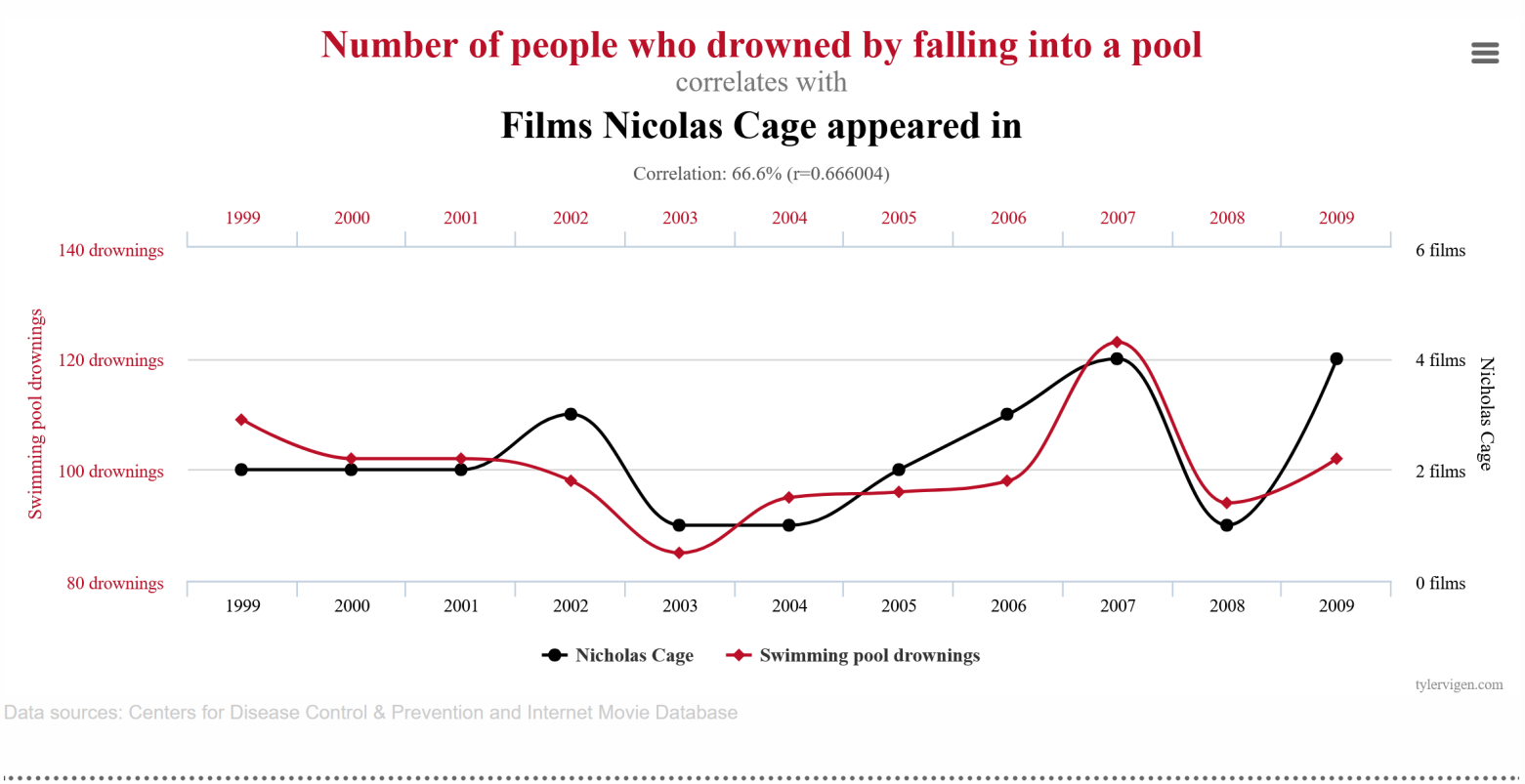
2. Plausible Theory



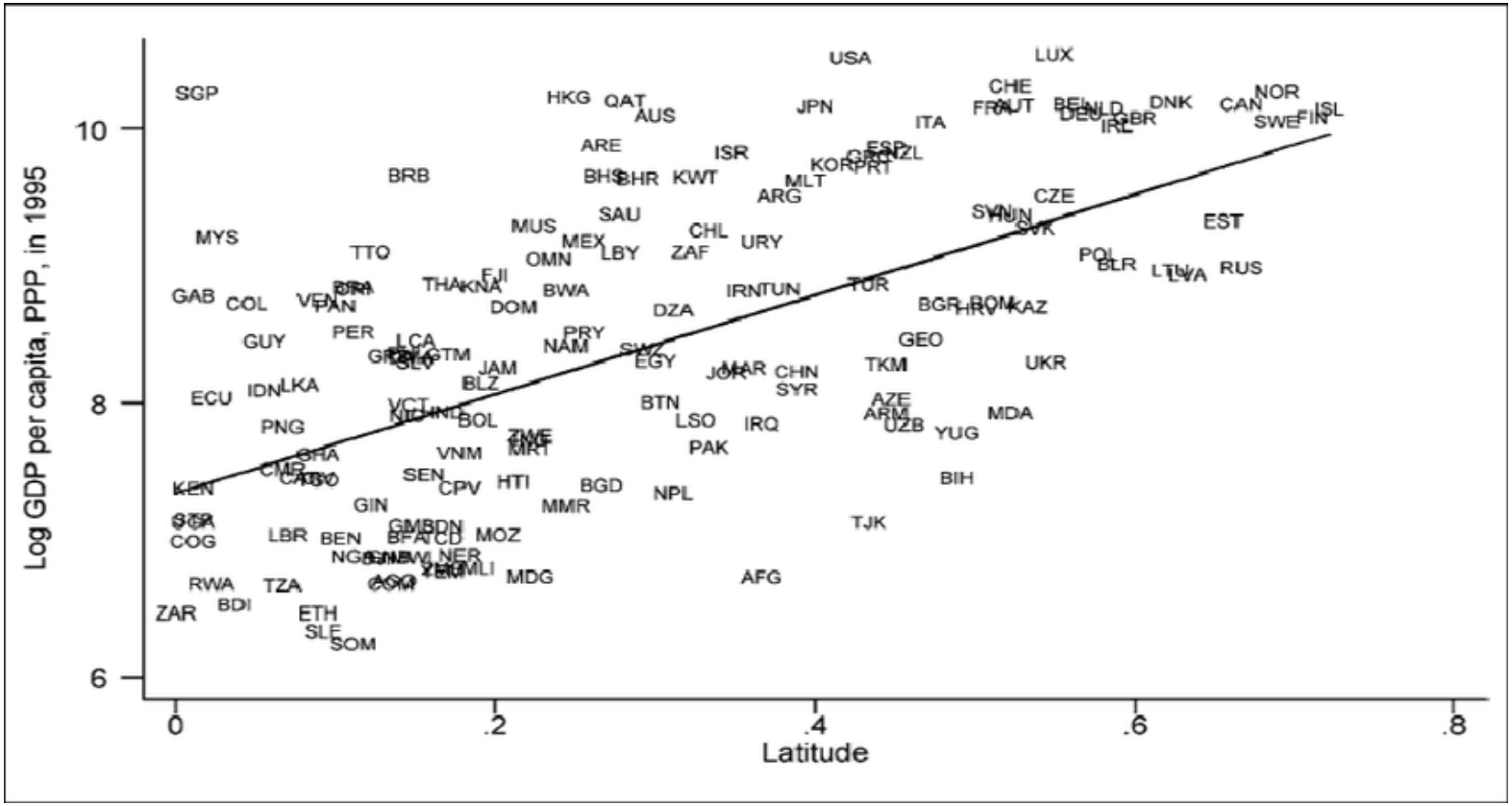
2. Plausible Theory



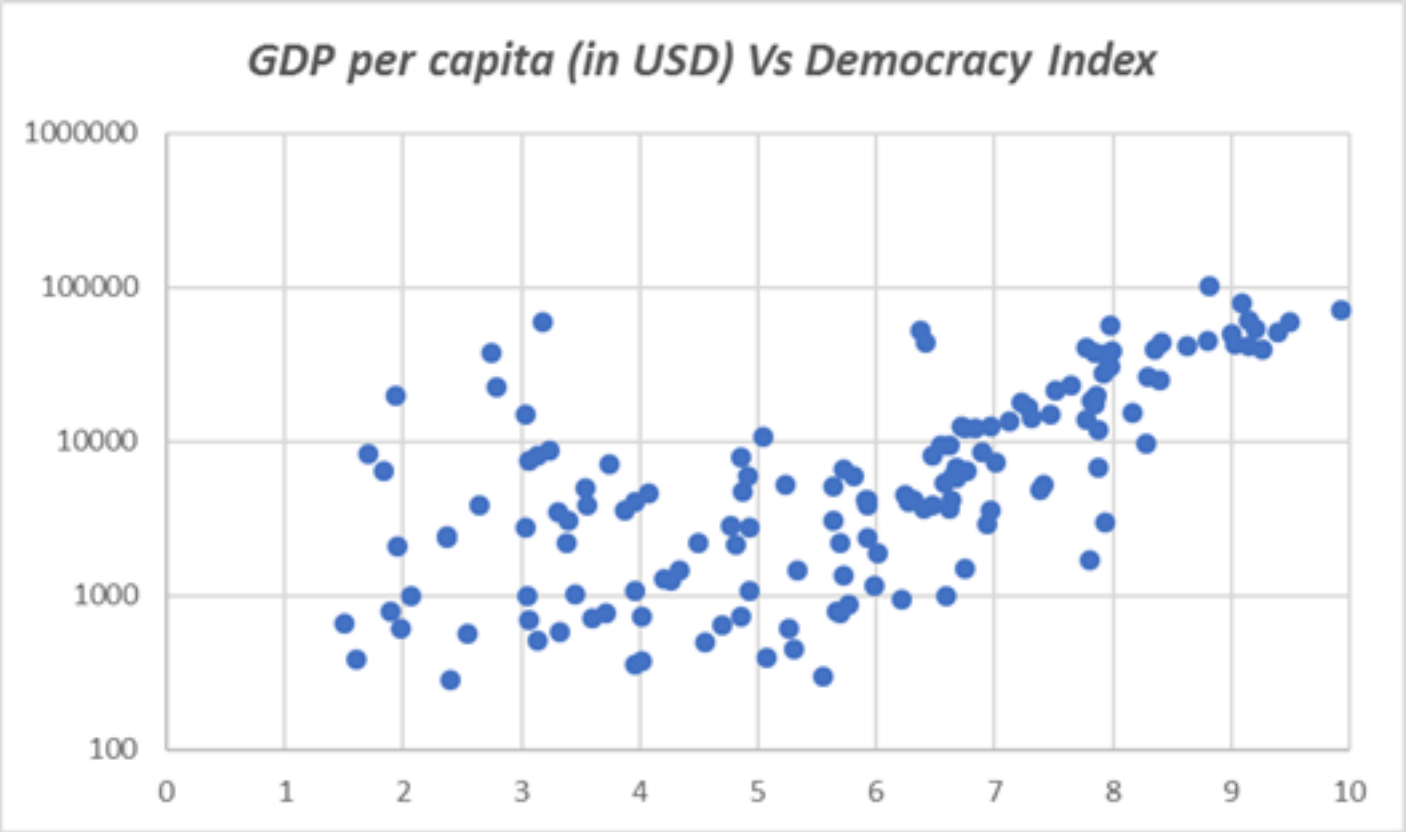
2. Plausible Theory



3. **Y** does not cause **X**



3. **Y** does not cause **X**



4. All Confounders are Ruled Out

Confounding Variable (**Z**)- Influences both the cause (**X**) and effect (**Y**)

Spurious Relationship- Correlation but without causation, due to coincidence or a confounding variable

- Again, ice cream sales and murders

4. All Confounders are Ruled Out

- Coke + Vodka = drunk

4. All Confounders are Ruled Out

- Coke + Vodka = drunk
- Coke + Rum = drunk

4. All Confounders are Ruled Out

- Coke + Vodka = drunk
- Coke + Rum = drunk
- Coke + Whiskey = drunk

4. All Confounders are Ruled Out

- Coke + Vodka = drunk
- Coke + Rum = drunk
- Coke + Whiskey = drunk

Soda makes people drunk!

Causal Relationships

1. There is <u>covariation</u> between **X** and **Y**
2. There is a plausible <u>causal mechanism</u>
3. **X** causes **Y**, <u>not the other way around</u> (temporal order)
4. <u>Confounders</u> are ruled out or controlled for

How do we test whether **X** causes **Y**?

Research Design

1. Experimental Studies- the researcher controls and randomly assigns values of the independent variable to the participant
   - The gold standard
2. Observational Studies- the researcher makes comparisons of units that already exist in the real world
   - What we do more often

Experiments

The researcher controls and randomly assigns values of the independent variable to the participant

1. The values of the independent variable the participants receive are determined by the researcher, not by the participants themselves or by nature
2. The values of the independent variable are *randomly assigned* to the participants

Estimate Treatment Effect

1. Measure average value of $\mathbf{Y}$ in Treatment Group ($\mathbf{X} = 1$)
2. Measure average value of $\mathbf{Y}$ in Control Group ($\mathbf{X} = 0$)
3. Calculate difference

The Magic of Random Assignment

Random Assignment- nothing systematic and substantively meaningful correlates with assignment to treatment or control group (or value of independent variable)

- Coin flips
- Random number generator
- Alphabetization?

The Magic of Random Assignment

Random Assignment- nothing systematic and substantively meaningful correlates with assignment to treatment or control group (or value of independent variable)

- Coin flips
- Random number generator
- Alphabetization?

If sample size is large enough, groups will be identical in the aggregate

- **Y** cannot be the cause of **X**
- Correlation between **X** and **Y** can be established
- Because groups are otherwise identical, **Z** is *accounted for...*

Types of Experiments

- Lab Experiments
- Survey Experiments
- Field Experiments
- Natural Experiments (Technically not an experiment, but close)

If experiments are so great, why not always do them?

Observational Studies

The researcher makes comparisons of units that already exist in the real world.

- Researcher does not have control over independent variable assignment
- *Approximate* experiment by explicitly accounting for confounders
- Often more practical when random assignment is impossible, too costly, or unethical

Observational Studies

The researcher makes comparisons of units that already exist in the real world.

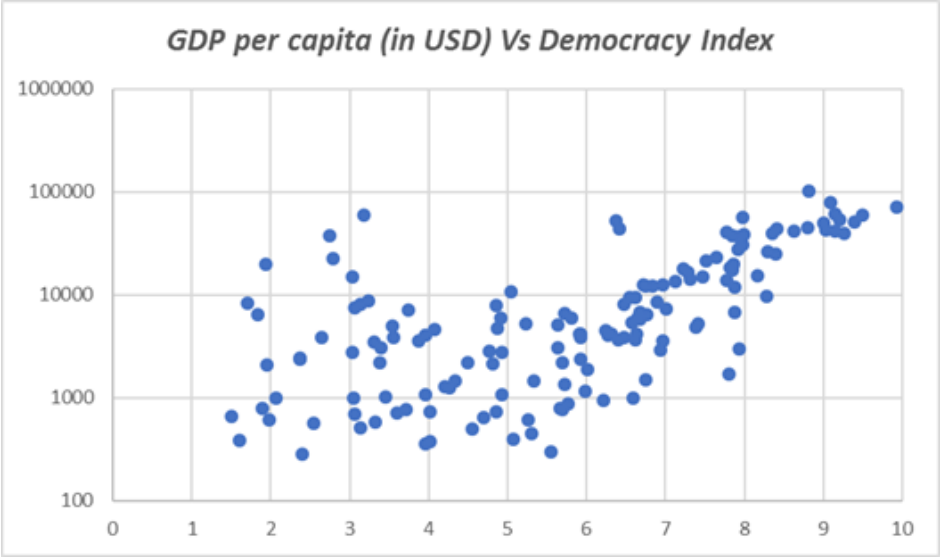
- Researcher does not have control over independent variable assignment
- *Approximate* experiment by explicitly accounting for confounders
- Often more practical when random assignment is impossible, too costly, or unethical

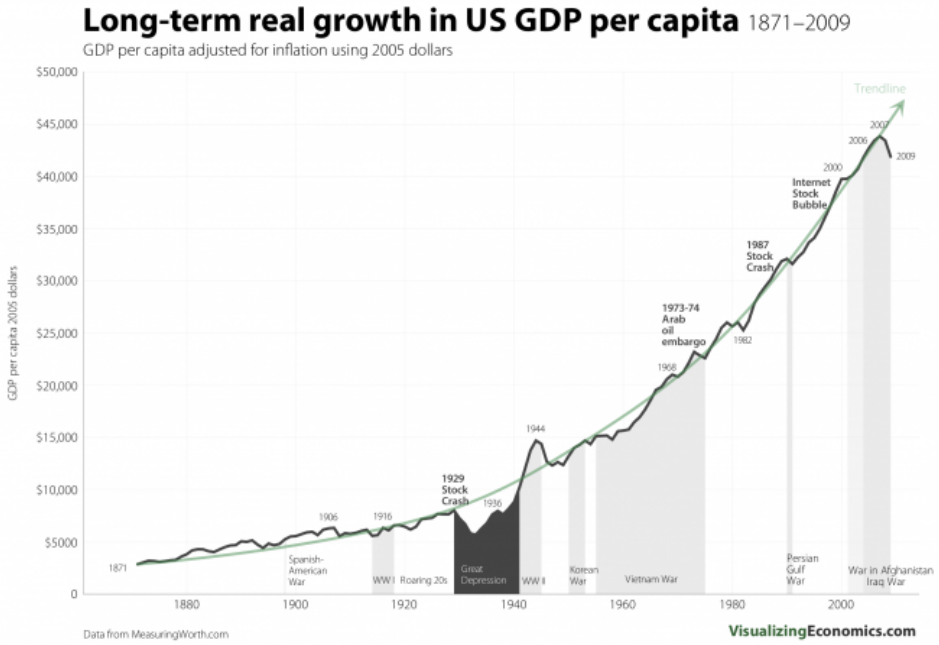
Can we establish causation with observational studies?

Data Structures

Cross-sectional- Compare values for variables across spatial or individual units at one point in time

Time-series- Compare values for variables across time with just one subject





Evaluating Research Designs

1. To what extent does it establish a causal relationship?

2. Internal Validity- Are we confident in our conclusions with respect to the specific subjects being studied?

3. External Validity- Can we generalize our conclusions to the broader population?
   - External sample validity- does our sample represent the broader population?
   - External validity of stimulus- is the treatment capturing the effect it would actually pick up in the real world?

Evaluating Research Designs

In general, experiments are high in internal validity but low in external validity.

But both types of studies can face threats to both types of validity.

- Threats to Internal Validity (a few examples)
  - Weird issues that pop up between measurements
  - Pretest effects
  - Mortality
- Threats to External Validity (a few examples)
  - Poor representation in sampling
  - Testing effects and Reactivity

Next Class

- Measurement:
  1. Read KW 5.1-5.3 (pages 104-115)
  2. Skim Eck and Fariss (2018)
- Start thinking about a research topic.